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MULTINOMIAL APPROXIMATING MODELS FOR OPTIONS WITH k STATE VARIABLES*

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Contingent claims whose values depend on multiple sources of uncertainty arise in many financial contracts and in the analysis of real projects. Unfortunately closed form solutions for these options are rare and numerical methods can be computationally expensive. This article extends the literature on multinomial approximating models. Specifically, new multinomial models are presented that include as special cases existing models. The more general models are shown to be computationally more efficient.

(CONTINGENT CLAIMS; OPTION PRICING; GEOMETRIC WIENER PROCESSES; MULTINOMIAL LATTICE)

1. Introduction

Contingent claims whose values depend on multiple sources of uncertainty arise in many financial contracts and in the analysis of real projects.¹ Unfortunately closed form solutions are rare, and in many cases numerical approximation procedures must be used. One common method is based on approximating the partial differential equation with a system of difference equations.² An alternative method is to approximate the underlying stochastic processes of the state variables. The binomial lattice approach of Cox, Ross and Rubinstein (1979) fits into this class, and is attractive because it explicitly establishes the arbitrage strategy that replicates the claim. However, providing a direct link between the approximating process and the arbitrage strategy is not essential. Specifically, if the market is complete and trading strategies allow no arbitrage opportunities, then an equivalent martingale measure exists that allows any contingent claim to be valued by computing an appropriate conditional expectation.³ If this conditional expectation is difficult to evaluate, then approximating processes can be taken and numerical methods invoked. Recently, Boyle (1988) used a multiperiod trinomial procedure to approximate a risk neutralized Geometric Wiener process. In Boyle's model the lattice jump probabilities were obtained by equating the first two moments of the underlying lognormal distribution to those of the approximating distribution. To ensure all jump probabilities were non-negative, Boyle introduced a stretch parameter, λ , that had to be constrained. Given a feasible stretch parameter, the valuation of the conditional expectation is accomplished by backward recursion.

Boyle also establishes a five-jump model to approximate a joint bivariate lognormal process and shows its use by valuing American style options that depend on prices of two state variables. Unfortunately, extending this procedure to three or more state variables is cumbersome because sets of parameters yielding nonnegative probabilities for the jumps have to be first obtained, and the difficulty of selecting suitable parameters results in

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¹ For a few examples of such claims see Boyle (1988), Omberg (1988), Johnson (1987), Stulz (1982) and Mason and Merton (1985).

² Brennan and Schwartz (1977, 1978) and Courtadon (1982).

³ See Cox and Ross (1976) and Harrison and Pliska (1981) for discussions on this point.

implementation problems. To overcome these problems Boyle, Evnine and Gibbs (BEG) (1989) considered an alternative approximating procedure. Specifically, for the two-state variable problem they used a 4-jump multiperiod lattice to approximate the *logarithmic return* process. By equating the moment generating function of the approximating distribution to the true normal moment generating function, in each time increment, a system of four equations in four unknowns was obtained. The unique probability expressions for their multinomial distribution were then used to value American style claims. The approach is quite elegant since it overcomes the problem of the earlier model and generalizes readily to k state variables.

The purpose of this article is to develop an alternative approximating technique for valuing claims on one or more state variables. Specifically, we develop a multinomial approximating option model that has advantages over existing models. Like the BEG model, our model is based on approximating the logarithmic returns process by a discrete multinomial lattice. Unlike the BEG model, however, we allow for horizontal jumps. We show that within our framework, the binomial option model appears as a special case of our one-state variable model, and the BEG model appears as a special case of our two-state model.

In the next section we introduce the one-state option pricing model and compare its performance to the well-known binomial model. Then we develop the two-state option model and compare it to the BEG model. Finally we generate the k -state model, and illustrate it by investigating a 3-state model. Throughout the paper we assume the setting of Black and Scholes (1973) model.

2. The Single-State Model

Assume the underlying asset follows a Geometric Wiener Process which, for valuation purposes has drift $\mu = r - \sigma^2/2$, where r is the riskless rate and σ is the instantaneous volatility. Then

TABLE 1
Difference Between True and Computed Option Prices

Strike X	λ	p_2	$n = \text{Number of Iterations}$						
			10	20	30	40	50	60	70
35.0	1.41421	0.500	0.015	-0.001	-0.002	0.003	-0.001	0.000	0.001
	1.29100	0.400	0.005	0.007	-0.001	0.000	0.003	0.000	0.000
	1.22474	0.333	-0.003	0.001	0.003	-0.002	0.005	0.000	0.004
	1.11803	0.200	-0.009	-0.004	-0.001	0.002	-0.001	0.002	-0.001
	1.05409	0.100	-0.009	-0.003	-0.002	0.000	0.001	-0.001	0.001
	1.000	0.000	-0.019	0.013	-0.006	-0.002	0.005	-0.003	-0.002
40.0	1.41421	0.500	0.030	0.010	0.007	0.004	0.003	0.002	0.000
	1.29100	0.400	0.018	0.007	0.004	0.002	0.001	0.000	0.000
	1.22474	0.333	0.013	0.005	0.002	0.001	0.000	-0.001	-0.001
	1.11803	0.200	0.006	0.001	0.000	-0.001	-0.002	-0.002	-0.002
	1.05409	0.100	0.008	0.000	-0.001	-0.002	-0.003	-0.003	-0.003
	1.000	0.000	0.056	0.027	0.017	0.011	0.008	0.006	0.005
45.0	1.41421	0.500	-0.011	-0.011	0.008	-0.007	-0.001	-0.003	-0.005
	1.29100	0.400	0.009	-0.008	-0.004	-0.003	-0.003	-0.005	0.001
	1.22474	0.333	0.023	-0.002	-0.007	0.003	-0.005	-0.001	-0.003
	1.11803	0.200	0.001	0.003	-0.003	-0.005	-0.001	-0.004	-0.003
	1.05409	0.100	-0.005	-0.005	0.004	-0.005	-0.002	-0.004	-0.002
	1.000	0.000	0.000	-0.012	0.007	-0.011	-0.006	0.008	-0.003

Case parameters: $S(0) = 40.0$; $\sigma = 0.20$; $r = 4.879\%$; $T = 7$ months (0.58333 years).

$$\ln \{S(t + \Delta t)\} = \ln \{S(t)\} + \zeta(t) \quad (1)$$

where $\zeta(t)$ is a normal random variable with mean $\mu\Delta t$ and variance $\sigma^2\Delta t$.

Let $\zeta^a(t)$ be the approximating distribution for $\zeta(t)$ over the period $[t, t + \Delta t]$. $\zeta^a(t)$ is a discrete random variable with the following distribution

$$\zeta^a(t) = \begin{cases} v & \text{with probability } p_1, \\ 0 & \text{with probability } p_2, \\ -v & \text{with probability } p_3, \end{cases}$$

where $v = \lambda\sigma\sqrt{\Delta t}$, and $\lambda \geq 1$.

The mean and variance of the approximating distribution are chosen to equal the mean and variance of $\zeta(t)$. Specifically, we have:

$$E \{ \zeta^a(t) \} = v(p_1 - p_3) = \mu\Delta t, \quad (2a)$$

$$\text{Var} \{ \zeta^a(t) \} = v^2(p_1 + p_3) = \sigma^2\Delta t + O(\Delta t). \quad (2b)$$

Substituting $v = \lambda\sigma\sqrt{\Delta t}$, and for sufficiently small Δt , while recognizing $\sum_{j=1}^3 p_j = 1$ yields

$$p_1 = \frac{1}{2\lambda^2} + \frac{\mu\sqrt{\Delta t}}{2\lambda\sigma}, \quad (3a)$$

$$p_2 = 1 - 1/\lambda^2, \quad (3b)$$

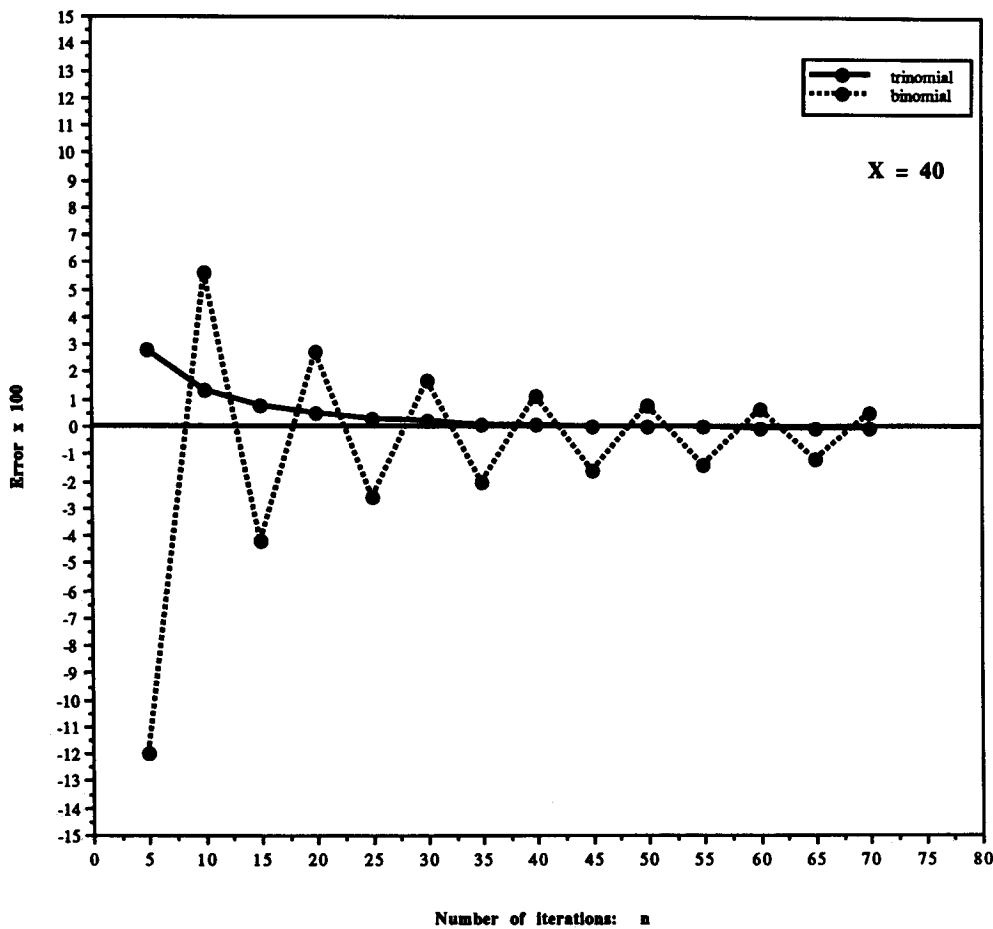


FIGURE 1. Convergence Rates of Trinomial and Binomial Models for an At-the-Money Call Option.

$$p_3 = \frac{1}{2\lambda^2} - \frac{\mu\sqrt{\Delta t}}{2\lambda\sigma}. \quad (3c)$$

By choosing different values for λ , a range of values for the above probabilities can be obtained through which option valuation can be accomplished. Note that if $\lambda = 1$, $p_2 = 0$, and the expressions collapse to the binomial model of Cox, Ross and Rubinstein (1979). Note too, that unlike Boyle's trinomial procedure, any value for λ , $\lambda \geq 1$, yields a feasible set of probabilities.

Table 1 illustrates the difference between true (Black-Scholes) and computed European call prices for different trinomial specifications for in, at, and, out-the-money contracts.

Notice that when λ is chosen such that the horizontal jump probably is one third, the error terms are generally quite small. For this reason, in what follows we restrict attention to the setting of λ , and compare the results to the binomial model. Figure 1 compares the speed of convergence for an at-the-money European call option for the case $\lambda = 1.22474$ (which corresponds to $p_2 = \frac{1}{3}$) and $\lambda = 1$ (which corresponds to $p_2 = 0$ or the binomial model).

Figure 1 shows that the results for the trinomial model after 15 iterations are comparable to the results of the binomial model after 55 iterations. The results for in and out-the-money options are somewhat similar. See Figures 2.1 and 2.2.

Of course a trinomial iteration is computationally more expensive, so unless the

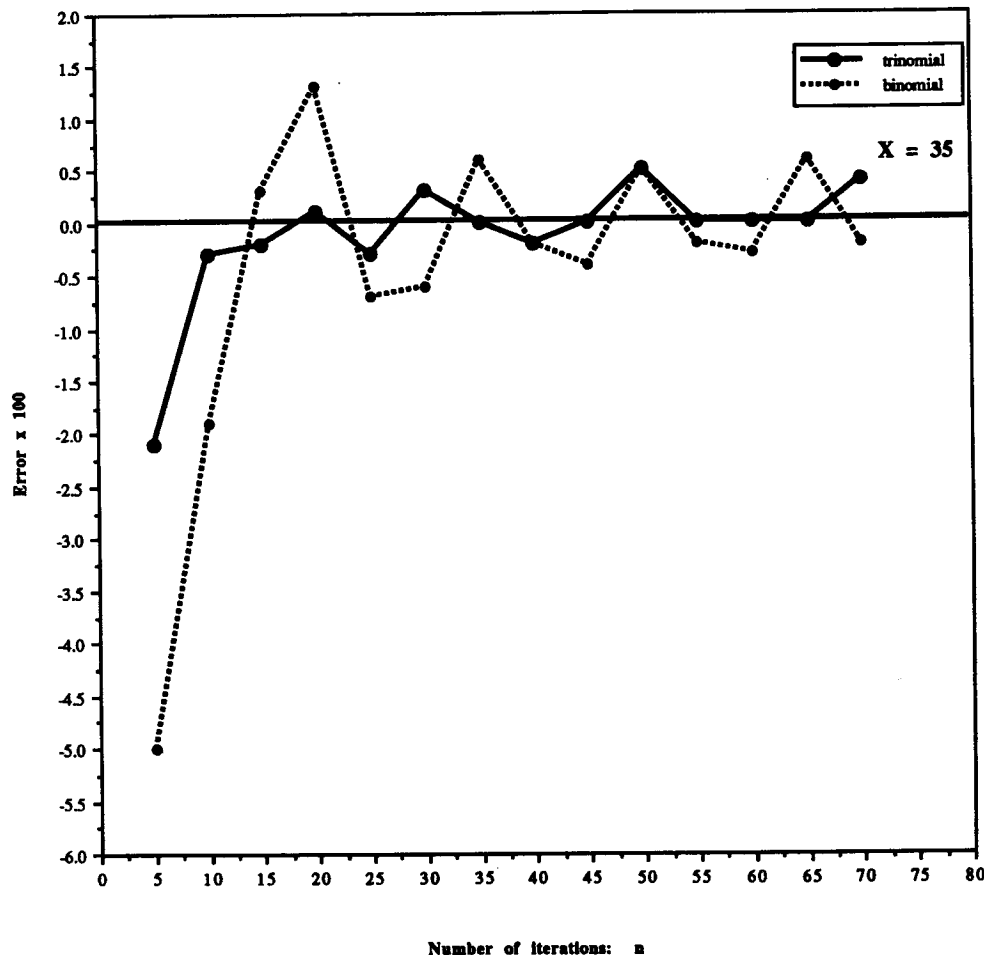


FIGURE 2.1. Convergence Rates of Trinomial and Binomial Models for an In-the-Money Call Option.

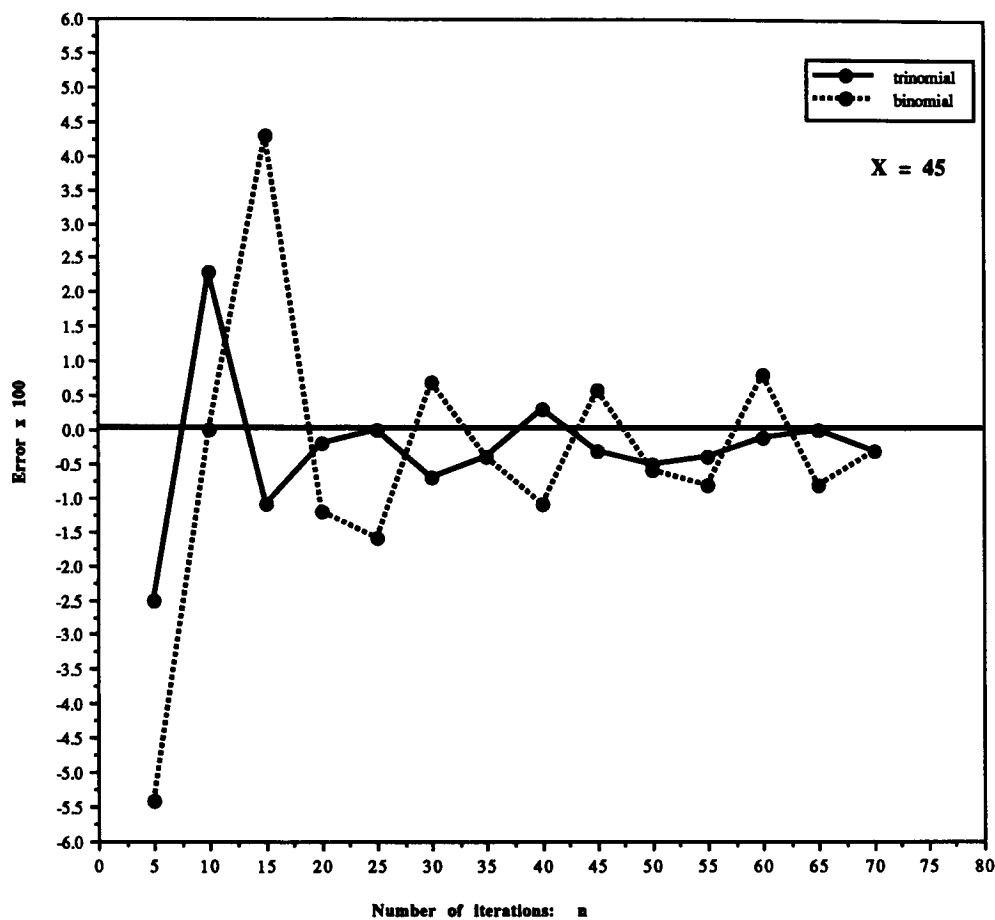


FIGURE 2.2. Convergence Rates of Trinomial and Binomial Models for an Out-the-Money Call Option.

convergence rate is significantly more rapid, the advantage disappears. Let $N_k(n)$ be the total number of nodes generated in n iterations for a model with k state variables, then

$$N_1(n) = \begin{cases} \sum_{i=0}^n (2i + 1) = (n + 1)^2 & \text{for a trinomial model,} \\ \sum_{i=0}^n (i + 1) = (n + 1)(n + 2)/2 & \text{for a binomial model.} \end{cases}$$

Except for the last period, where option values are determined by their boundary conditions, each node in the trinomial model requires 3 multiplications and 2 additions, while the binomial model requires 2 multiplications and 1 addition.⁴

Hence, the total number of multiplications and additions are:

Model	No. of Multiplications	No. of Additions
trinomial	$3n^2$	$2n^2$
binomial	$n^2 + n$	$\frac{1}{2}(n^2 + n)$

Clearly a trinomial model with n steps requires less computational work (measured in terms of multiplications and additions) than a binomial model with $2n$ steps. Table

⁴ Here, a node refers to a possible state generated by the approximating process. For instance a binomial process, after an elapsed time of length Δt , generates two possible nodes (states). An up node and a down node.

TABLE 2
Comparison of Errors Between the Trinomial Model Using n Iterations
and the Binomial Model with $2n$ Iterations

n Iterations on a Trinomial	Error on Trinomial with n Iterations	Error on Binomial with $2n$ Iterations*
10	0.013	0.027
20	0.005	0.011
30	0.002	0.006
40	0.001	0.004
50	0.000	0.002

(Table 1 case parameters are used).

* The total amount of multiplications and additions on a binomial with $2n$ iterations exceeds that of the trinomial with n iterations.

2 compares the accuracy of the two models for pricing at-the-money options on a work-adjusted basis. In all cases the error of the trinomial model was at least half the size of the binomial model error. Table 3 compares the accuracy of the trinomial model with the binomial model over a wide range of parameters when 20 iterations are used for the trinomial model and 40 iterations for the binomial model. In almost all cases the trinomial model was more accurate.

The benefits of this trinomial approximation scheme are now extended to two sources of uncertainty.

3. Two-State Models

Let $\{S_1(t), S_2(t)\}$ define the asset price pair at time t . Assume the joint density of the two underlying securities is bivariate lognormal. Let $\mu_i = r - \sigma_i^2/2$ and σ_i^2 be the instantaneous mean and variance for asset i ($i = 1, 2$) and let ρ be the correlation coefficient. As before, for each asset we have over $[t, t + \Delta t]$.

$$\ln S_i(t + \Delta t) = \ln S_i(t) + \zeta_i(t)$$

where $\zeta_i(t)$ is a normal random variable with mean $\mu_i \Delta t$ and variance $\sigma_i^2 \Delta t$. The instantaneous correlation between $\zeta_1(t)$ and $\zeta_2(t)$ is ρ .

TABLE 3
Comparison of Errors; Trinomial ($n = 20$) with Binomial ($n = 40$)

		Maturity (months)					
		1		4		7	
Volatility	Strike X	Trinomial	Binomial	Trinomial	Binomial	Trinomial	Binomial
$\sigma = 0.20$	35	0.002	0.002	0.000	0.003	0.001	0.002
	40	0.000	0.003	0.010	0.014	0.005	0.011
	45	0.001	0.002	0.000	0.005	0.002	0.011
$\sigma = 0.30$	35	0.001	0.003	0.003	0.009	0.001	0.002
	40	0.003	0.007	0.006	0.014	0.016	0.027
	45	0.003	0.002	0.015	0.001	0.006	0.012
$\sigma = 0.40$	35	0.003	0.000	0.003	0.006	0.030	0.025
	40	0.006	0.011	0.010	0.024	0.016	0.030
	45	0.000	0.002	0.004	0.020	0.020	0.006

Case parameters: $S(0) = 40.0$, $r = 4.879\%$.

The joint normal random variable $\{\zeta_1(t), \zeta_2(t)\}$ will be approximated by a pair of multinomial discrete random variables having the following distribution.

$\zeta_1^a(t)$	$\zeta_2^a(t)$	Probability
v_1	v_2	p_1
v_1	$-v_2$	p_2
$-v_1$	$-v_2$	p_3
$-v_1$	v_2	p_4
0	0	p_5

where $v_i = \lambda_i \sigma_i \sqrt{\Delta t}$ ($i = 1, 2$).

In order to ensure the convergence of the approximating distribution to the true distribution as $\Delta t \rightarrow 0$ the first two moments of the approximating distribution are set equal to the true moments of the continuous distribution. Specifically, we have

$$E\{\zeta_1^a(t)\} = v_1(p_1 + p_2 - p_3 - p_4) = \mu_1 \Delta t, \quad (4a)$$

$$E\{\zeta_2^a(t)\} = v_2(p_1 - p_2 - p_3 + p_4) = \mu_2 \Delta t, \quad (4b)$$

$$\text{Var}\{\zeta_1^a(t)\} = v_1^2(p_1 + p_2 + p_3 + p_4) = \sigma_1^2 \Delta t + O(\Delta t), \quad (4c)$$

$$\text{Var}\{\zeta_2^a(t)\} = v_2^2(p_1 + p_2 + p_3 + p_4) = \sigma_2^2 \Delta t + O(\Delta t). \quad (4d)$$

In addition to the equality of the two moments, the covariance terms must also be equal. This can be done by equating the expected value of the product of the two variables which yields

$$E\{\zeta_1^a(t)\zeta_2^a(t)\} = v_1 v_2(p_1 - p_2 + p_3 - p_4) = \sigma_1 \sigma_2 \rho \Delta t + O(\Delta t). \quad (4e)$$

Substituting $v_i = \lambda_i \sigma_i \sqrt{\Delta t}$ ($i = 1, 2$), and for sufficiently small Δt , the above equations simplify to

TABLE 4
Difference Between True and Computed Option Prices on the Maximum of Two Assets

Strike X	λ	p_5	$n = \text{Number of Iterations}$						
			10	20	30	40	50	60	70
35.0	1.41421	0.500	0.085	0.028	0.015	0.011	0.009	0.007	0.006
	1.29100	0.400	0.052	0.026	0.009	0.008	0.006	0.004	0.004
	1.22474	0.333	0.030	0.013	0.008	0.006	0.004	0.003	0.004
	1.11803	0.200	0.018	0.006	0.005	0.004	0.002	0.004	0.002
	1.05409	0.100	0.012	0.003	0.003	0.001	0.003	0.001	0.002
	1.000	0.000	0.020	0.012	-0.002	0.003	0.008	0.001	0.000
40.0	1.41421	0.500	0.078	0.037	0.025	0.018	0.015	0.012	0.011
	1.29100	0.400	0.053	0.026	0.018	0.013	0.011	0.009	0.008
	1.22474	0.333	0.037	0.019	0.013	0.009	0.008	0.007	0.005
	1.11803	0.200	0.026	0.013	0.009	0.007	0.005	0.005	0.004
	1.05409	0.100	0.025	0.009	0.006	0.005	0.004	0.003	0.003
	1.000	0.000	0.087	0.044	0.030	0.022	0.0180	0.015	0.013
45.0	1.41421	0.500	0.003	-0.010	0.024	0.000	0.002	0.005	0.006
	1.29100	0.400	-0.006	-0.001	0.004	-0.001	0.004	0.004	0.003
	1.22474	0.333	-0.009	0.013	-0.003	0.003	0.005	0.003	-0.001
	1.11803	0.200	-0.020	0.012	-0.003	0.003	0.002	-0.002	0.001
	1.05409	0.100	-0.021	0.000	0.002	0.002	-0.001	-0.001	0.002
	1.000	0.000	0.000	0.006	-0.004	-0.012	0.001	0.019	0.002

Case parameters: $S_1(0) = S_2(0) = 40.0$; $\sigma_1 = 0.20$, $\sigma_2 = 0.30$; $\rho = 0.50$; $r = 4.879\%$; $T = 7$ months (0.58333 years).

$$p_1 + p_2 - p_3 - p_4 = \frac{\mu_1 \sqrt{\Delta t}}{\lambda_1 \sigma_1}, \quad (5a)$$

$$p_1 - p_2 - p_3 + p_4 = \frac{\mu_2 \sqrt{\Delta t}}{\lambda_2 \sigma_2}, \quad (5b)$$

$$p_1 + p_2 + p_3 + p_4 = \frac{1}{\lambda_1^2}, \quad (5c)$$

$$p_1 + p_2 + p_3 + p_4 = \frac{1}{\lambda_2^2}, \quad (5d)$$

$$p_1 - p_2 + p_3 - p_4 = \frac{\rho}{\lambda_1 \lambda_2}. \quad (5e)$$

Equations (5c) and (5d) imply that $\lambda_1 = \lambda_2 = \lambda$ with $\lambda \geq 1$. The resulting four equations can be solved in terms of λ to yield expressions for p_1 , p_2 , p_3 , and p_4 . Since $\sum_{j=1}^5 p_j = 1$, then the solution is

$$p_1 = \frac{1}{4} \left\{ \frac{1}{\lambda^2} + \frac{\sqrt{\Delta t}}{\lambda} \left(\frac{\mu_1}{\sigma_1} + \frac{\mu_2}{\sigma_2} \right) + \frac{\rho}{\lambda^2} \right\}, \quad (6a)$$

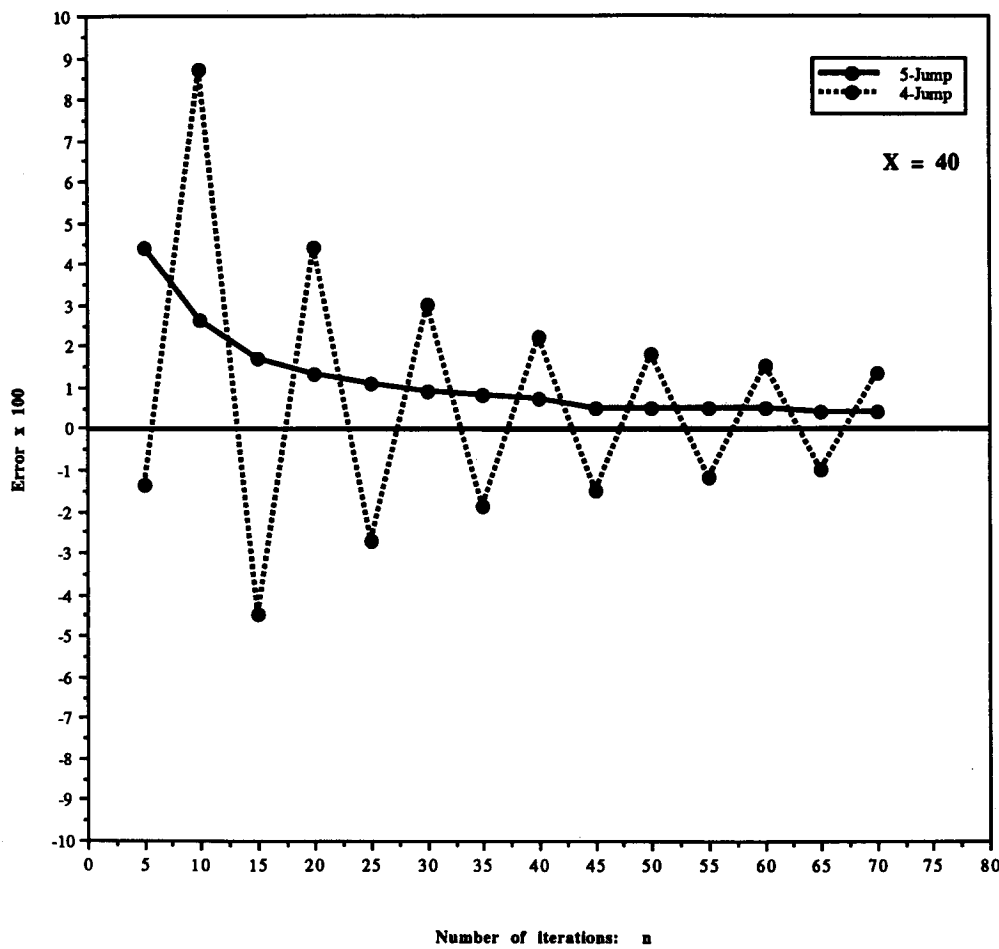


FIGURE 3. Convergence Rates of 5-Jump and 4-Jump Models for an At-the-Money Call Option on the Maximum of Two Assets.

$$p_2 = \frac{1}{4} \left\{ \frac{1}{\lambda^2} + \frac{\sqrt{\Delta t}}{\lambda} \left(\frac{\mu_1}{\sigma_1} - \frac{\mu_2}{\sigma_2} \right) - \frac{\rho}{\lambda^2} \right\}, \quad (6b)$$

$$p_3 = \frac{1}{4} \left\{ \frac{1}{\lambda^2} + \frac{\sqrt{\Delta t}}{\lambda} \left(-\frac{\mu_1}{\sigma_1} - \frac{\mu_2}{\sigma_2} \right) + \frac{\rho}{\lambda^2} \right\}, \quad (6c)$$

$$p_4 = \frac{1}{4} \left\{ \frac{1}{\lambda^2} + \frac{\sqrt{\Delta t}}{\lambda} \left(-\frac{\mu_1}{\sigma_1} + \frac{\mu_2}{\sigma_2} \right) - \frac{\rho}{\lambda^2} \right\}, \quad (6d)$$

$$p_5 = 1 - \frac{1}{\lambda^2}. \quad (6e)$$

Unlike Boyle's 5-jump model, the search for a set of nonnegative probability values is not necessary. Indeed any value of λ , $\lambda \geq 1$, will suffice. Note, too, that if $\lambda = 1$, $p_5 = 0$, and the overall approximation process reduces to the four-jump process obtained by the BEG model.

To examine the efficiency of this approach we investigate how well it approximates the price of an option on the maximum of two securities. Such options have been well studied, and Stulz (1982) has provided analytical solutions for them. Table 4 shows the difference between the true and computed European Call option prices on the maximum

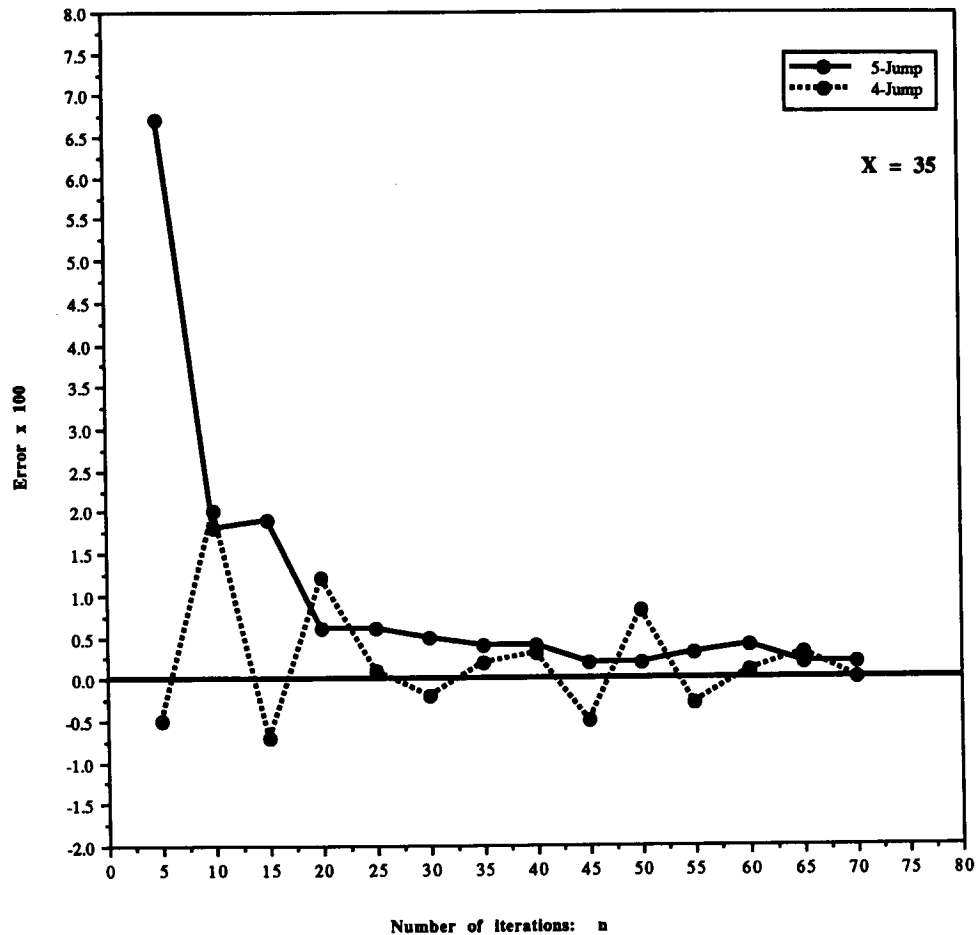


FIGURE 4.1. Convergence Rates of 5-Jump and 4-Jump Models for an In-the-Money Call Option on the Maximum of Two Assets.

of two assets for in, at, and out-the-money contracts, for different λ values, and for different numbers of iterations.

Notice that for values of λ that force p_5 to be near $\frac{1}{5}$, the results are quite satisfactory, while for very small ($p_5 = 0$) and large ($p_5 = 0.50$) values the residual terms are large.

Figure 3 compares the rates of convergence of the 5-jump model with $\lambda = 1.11803$ (corresponding to $p_5 = \frac{1}{5}$) to the case where $\lambda = 1$ (which corresponds to $p_5 = 0$, or the BEG model) for an at-the-money European call option on the maximum of two securities.

As can be seen, the results are somewhat similar to those of Figure 1. That is the speed of convergence is smoother and more rapid for the 5-jump model. Figures 4.1 and 4.2 show that similar results hold for in and out-the-money options.

For approximating models involving two state variables the total number of nodes generated by the process after n iterations is

$$N_2(n) = \begin{cases} \sum_{i=0}^n [(i+1)^2 + i^2] = (4n^3 + 12n^2 + 14n + 6)/6 & \text{for a 5-jump model,} \\ \sum_{i=0}^n (i+1)^2 = (2n^3 + 9n^2 + 13n + 6)/6 & \text{for a 4-jump model.} \end{cases}$$

Except for the last period, where the option values are governed by the boundary

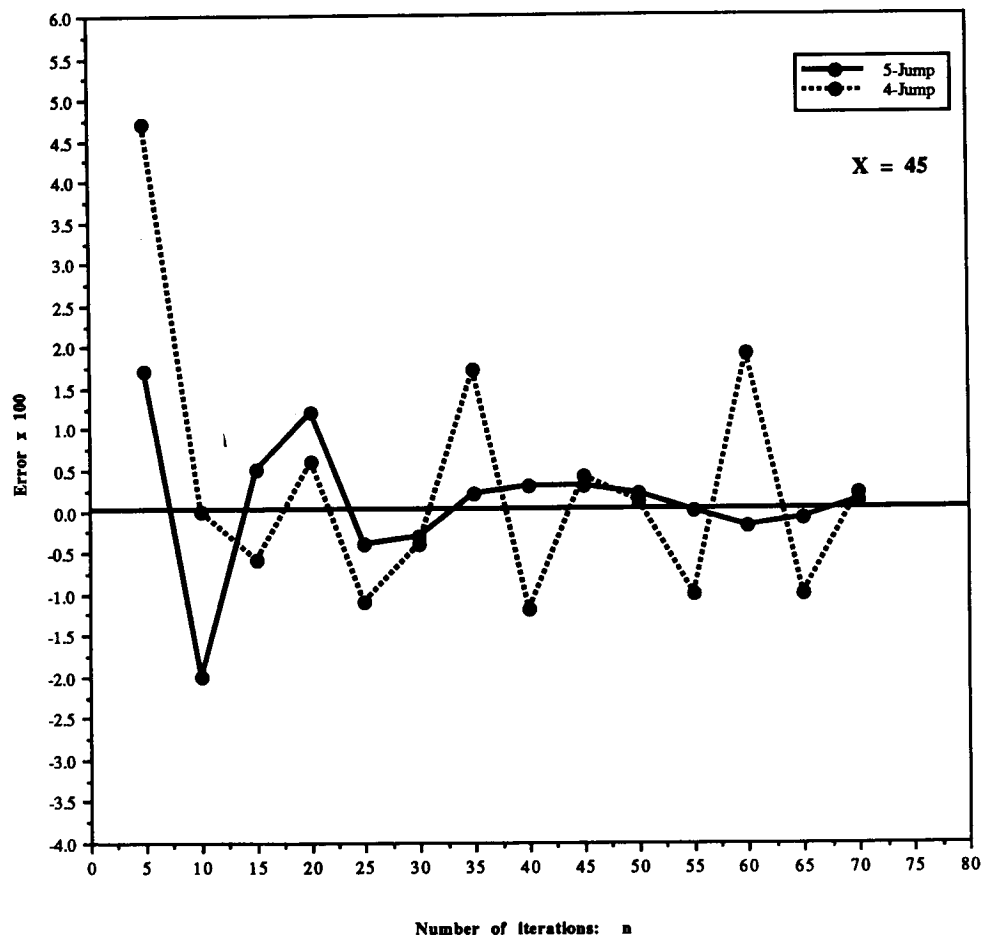


FIGURE 4.2. Convergence Rates of 5-Jump and 4-Jump Models for an Out-the-Money Call Option on the Maximum of Two Assets.

TABLE 5
Comparison of Errors Between the 5-Jump Model Using n Iterations and 4-Jump Model Using n' Iterations

n Iterations on the 5 Jump	n' Iterations on the 4 Jump	Error on	
		5 Jump with n Iterations	4 Jump with n' Iterations*
10	15	0.026	-0.045
20	30	0.013	0.030
30	45	0.009	-0.015
40	55	0.007	-0.012
50	70	0.005	0.013

(Table 2 case parameters are used.)

* The total amount of multiplications and additions on a 4-jump model with n' iterations exceeds that of a 5-jump model with n iterations.

conditions, each node in a 5-jump model requires five multiplications and four additions, while the 4-jump model requires one less multiplication and one less addition per given node. Therefore, the total number of multiplications and additions are:

Model	No. of Multiplications	No. of Additions
5-jump	$\frac{5}{3}(2n^3 + n)$	$\frac{4}{3}(2n^3 + n)$
4-jump	$\frac{3}{2}(2n^3 + 3n^2 + n)$	$\frac{1}{2}(2n^3 + 3n^2 + n)$

Table 5 compares the size of errors of a 4-jump model to a 5-jump model, where, in each case, the work done by the 4-jump model (measured in terms of multiplications and additions) dominates that of a 5-jump model.

In each case, the size of the error in the 4-jump model is approximately double that of the 5-jump model.

The 5-jump models developed to approximate the logarithmic return process here, avoid the problems of the 5-jump model that approximates the lognormally distributed prices of Boyle (1988), and, like the BEG model, can be readily extended to k sources of uncertainty.

4. Models with k Sources of Uncertainty

As before, it is assumed that the joint density of the asset prices follows a multivariate lognormal distribution. Let $\mu_i = r - \sigma_i^2/2$, and σ_i^2 be the instantaneous mean and variance, respectively ($i = 1, 2, \dots, k$) and let ρ_{ij} be the correlation between security i and j . When the approximation process includes the opportunity for horizontal jumps there exist $2^k + 1$ probability expressions such that:

$$\sum_{m=1}^{2^{k+1}} p_m = 1. \quad (7)$$

In order to ensure the existence of horizontal jumps set $v_i = \lambda_i \sigma_i \sqrt{\Delta t}$, with $\lambda_i \geq 1$ ($i = 1, 2, \dots, k$). Necessary conditions for convergence require the equality of the means, variances, and the pairwise covariance terms, where the variance equality equations provide the opportunity for setting $\lambda_i = \lambda \geq 1$ ($i = 1, \dots, k$). With k underlying sources of variability, our approximating process requires $2^k + 1$ jumps in an interval of length Δt . Let m define a given state of the process after an elapsed time of Δt , with p_m representing the probability of state m ($m = 1, 2, \dots, 2^k + 1$). Let $p_{2^{k+1}}$ represent the

probability of a horizontal jump, $k \geq 2$,⁵ that is, the probability that the prices at time t remain unchanged at time $(t + \Delta t)$. For the remaining 2^k probability terms we can show that:

$$p_m = \frac{1}{2^k} \left\{ \frac{1}{\lambda^2} + \frac{\sqrt{\Delta t}}{\lambda} \sum_{i=1}^k x_{im} \left(\frac{\mu_i}{\sigma_i} \right) + \frac{1}{\lambda^2} \sum_{i=1}^{k-1} \sum_{j=i+1}^k x_{ij}^m (\rho_{ij}) \right\},$$

$$m = 1, 2, \dots, 2^k, \quad k \geq 2, \quad \lambda \geq 1, \quad \text{where} \quad (8)$$

$$x_{im} = \begin{cases} 1 & \text{if asset } i \text{ has an up jump in state } m, \\ -1 & \text{if asset } i \text{ has a down jump in state } m \end{cases} \quad \text{and}$$

$$x_{ij}^m = \begin{cases} 1 & \text{if assets } i \text{ and } j \text{ have jumps in the same direction in state } m, \\ -1 & \text{if assets } i \text{ and } j \text{ have jumps in opposite direction in state } m. \end{cases}$$

Given equation (7), the probability expression for the horizontal jump, $p_{2^{k+1}}$ is

$$p_{2^{k+1}} = 1 - \frac{1}{\lambda^2}. \quad (9)$$

When $\lambda = 1$, the probability of a horizontal jump is zero, and equation (8) reduces to the BEG model (1989).

As an example of equations (8) and (9) consider the case of three sources of uncertainty. The approximating multivariate distribution, $\zeta_i^a(t)$ ($i = 1, 2, 3$) is then given by

$\zeta_1^a(t)$	$\zeta_2^a(t)$	$\zeta_3^a(t)$	Probability
v_1	v_2	v_3	p_1
v_1	v_2	$-v_3$	p_2
v_1	$-v_2$	v_3	p_3
v_1	$-v_2$	$-v_3$	p_4
$-v_1$	v_2	v_3	p_5
$-v_1$	v_2	$-v_3$	p_6
$-v_1$	$-v_2$	v_3	p_7
$-v_1$	$-v_2$	$-v_3$	p_8
0	0	0	p_9

The resulting expressions for p_j ($j = 1, \dots, 8$) from equation (8) reduce to

$$p_1 = \frac{1}{8} \left\{ \frac{1}{\lambda^2} + \frac{\sqrt{\Delta t}}{\lambda} \left(\frac{\mu_1}{\sigma_1} + \frac{\mu_2}{\sigma_2} + \frac{\mu_3}{\sigma_3} \right) + \frac{\rho_{12} + \rho_{13} + \rho_{23}}{\lambda^2} \right\},$$

$$p_2 = \frac{1}{8} \left\{ \frac{1}{\lambda^2} + \frac{\sqrt{\Delta t}}{\lambda} \left(\frac{\mu_1}{\sigma_1} + \frac{\mu_2}{\sigma_2} - \frac{\mu_3}{\sigma_3} \right) + \frac{\rho_{12} - \rho_{13} - \rho_{23}}{\lambda^2} \right\},$$

$$p_3 = \frac{1}{8} \left\{ \frac{1}{\lambda^2} + \frac{\sqrt{\Delta t}}{\lambda} \left(\frac{\mu_1}{\sigma_1} - \frac{\mu_2}{\sigma_2} + \frac{\mu_3}{\sigma_3} \right) + \frac{-\rho_{12} + \rho_{13} - \rho_{23}}{\lambda^2} \right\},$$

$$p_4 = \frac{1}{8} \left\{ \frac{1}{\lambda^2} + \frac{\sqrt{\Delta t}}{\lambda} \left(\frac{\mu_1}{\sigma_1} - \frac{\mu_2}{\sigma_2} - \frac{\mu_3}{\sigma_3} \right) + \frac{-\rho_{12} - \rho_{13} + \rho_{23}}{\lambda^2} \right\},$$

$$p_5 = \frac{1}{8} \left\{ \frac{1}{\lambda^2} + \frac{\sqrt{\Delta t}}{\lambda} \left(-\frac{\mu_1}{\sigma_1} + \frac{\mu_2}{\sigma_2} + \frac{\mu_3}{\sigma_3} \right) + \frac{-\rho_{12} - \rho_{13} + \rho_{23}}{\lambda^2} \right\},$$

⁵ For instance, when $k = 2$ and $m = 1$, p_m represents the probability of state 1 in which both assets have increased in value over the interval $[t, t + \Delta t]$, see §3.

$$\begin{aligned}
p_6 &= \frac{1}{8} \left\{ \frac{1}{\lambda^2} + \frac{\sqrt{\Delta t}}{\lambda} \left(-\frac{\mu_1}{\sigma_1} + \frac{\mu_2}{\sigma_2} - \frac{\mu_3}{\sigma_3} \right) + \frac{-\rho_{12} + \rho_{13} - \rho_{23}}{\lambda^2} \right\}, \\
p_7 &= \frac{1}{8} \left\{ \frac{1}{\lambda^2} + \frac{\sqrt{\Delta t}}{\lambda} \left(-\frac{\mu_1}{\sigma_1} - \frac{\mu_2}{\sigma_2} + \frac{\mu_3}{\sigma_3} \right) + \frac{\rho_{12} - \rho_{13} - \rho_{23}}{\lambda^2} \right\}, \\
p_8 &= \frac{1}{8} \left\{ \frac{1}{\lambda^2} + \frac{\sqrt{\Delta t}}{\lambda} \left(-\frac{\mu_1}{\sigma_1} - \frac{\mu_2}{\sigma_2} - \frac{\mu_3}{\sigma_3} \right) + \frac{\rho_{12} + \rho_{13} + \rho_{23}}{\lambda^2} \right\},
\end{aligned}$$

and from equation (9) we obtain $p_9 = 1 - 1/\lambda^2$.

When $\lambda = 1$, p_9 becomes zero and the resulting model reduces to the BEG model with three state variables.

5. Conclusion

The valuation of contingent claims whose values depend on multiple sources of uncertainty is an important problem in financial economics. Since numerical methods for valuing such claims can be computationally expensive, the need for efficient algorithms is clear. Approximating the underlying risk neutralized stochastic process by a multinomial lattice is known to produce good results.

An issue that is raised is what order multinomial process should be used to approximate joint normally distributed logarithmic returns. This article has shown that models that include a horizontal jump offer computational advantages.

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